

- 2 Water hyacinth, *Eichhornia crassipes*, is a plant that grows on the surface of waterways such as rivers and lakes and is an alien species in many parts of the world. The plant grows and reproduces to form dense mats. As a result, water hyacinth outcompetes local species and reduces biodiversity.

Weevils are a group of insects that feed on water hyacinth. Certain species of weevil are used in South Africa as part of a programme to control the spread of water hyacinth.

Researchers carried out a two-stage investigation to assess the effect of weevils on water hyacinth.

Stage 1 was a pilot experiment in the laboratory. Stage 2 was a field experiment.

Stage 1: Pilot experiment

- Water hyacinth plants were collected from the river and kept in a laboratory.
- Each plant was kept in a separate bucket of river water for 6 weeks.
- During the 6 weeks, the plants were inspected twice a day and any insects, including weevils, were removed.
- Feeding scars (damage caused by insect feeding) were counted.
- At the end of the 6 weeks, the buckets containing the plants were divided into 6 groups.
- The length of the longest leaf stalk (petiole) of each plant was measured.
- The number of feeding scars on the second youngest leaf (leaf 2) of each plant was counted.

Each group was then given a different treatment, as shown in Table 2.1.

Table 2.1

group	treatment	
1	not sprayed with insecticide	no weevils added
2	not sprayed with insecticide	1 mating pair of weevils added after 1 week
3	sprayed with insecticide	no weevils added
4	sprayed with insecticide	1 mating pair of weevils added 1 week after spraying with insecticide
5	sprayed with insecticide	1 mating pair of weevils added 2 weeks after spraying with insecticide
6	sprayed with insecticide	1 mating pair of weevils added 3 weeks after spraying with insecticide

After 4 weeks, the effects of the treatments on water hyacinth were assessed.

The results are shown in Table 2.2.

Table 2.2

group	mean length of longest petiole / cm $\pm$ s		mean number of feeding scars on leaf 2 $\pm$ s	
	before treatment	after treatment	before treatment	after treatment
1	41 $\pm$ 2.0	38 $\pm$ 3.0	23 $\pm$ 4.0	0
2	32 $\pm$ 3.5	29 $\pm$ 3.3	16 $\pm$ 3.0	22 $\pm$ 5.0
3	36 $\pm$ 2.0	35 $\pm$ 3.8	18 $\pm$ 4.0	0.5 $\pm$ 0.2
4	38 $\pm$ 2.0	36 $\pm$ 3.0	30 $\pm$ 9.0	0.5 $\pm$ 0.2
5	34 $\pm$ 2.5	31 $\pm$ 3.5	25 $\pm$ 5.0	0.5 $\pm$ 0.2
6	37 $\pm$ 3.8	35 $\pm$ 4.3	13 $\pm$ 3.5	1.0 $\pm$ 0.5

- (a) (i) Explain the reason for spraying the water hyacinth plants with insecticide in treatment 3.

.....

 [2]

- (ii) Explain why treatment 2 and treatments 4, 5 and 6 were included in the experiment.

treatment 2

 treatments 4, 5 and 6

 [2]

- (iii) Explain what the standard deviation (s) shows about the results in Table 2.2.

.....

 [2]

- (iv) State **one** conclusion that can be made from the results in Table 2.2.

.....
 [1]

Stage 2: Field experiment

- 10 plots, each of 20 m², were marked out in a river.
- The plots were filled with water hyacinth plants.
- The plants were sampled at the start and the mean petiole length and number of feeding scars were recorded, as in the pilot experiment.
- 5 of the plots were sprayed with insecticide every three weeks.
- 5 of the plots were not sprayed, to allow weevils from the environment to feed on the plants.
- The plants were sampled after 33 weeks and the mean petiole length and number of feeding scars were recorded, as in the pilot experiment.

The results of the field experiment are shown in Fig. 2.1 and Fig. 2.2.

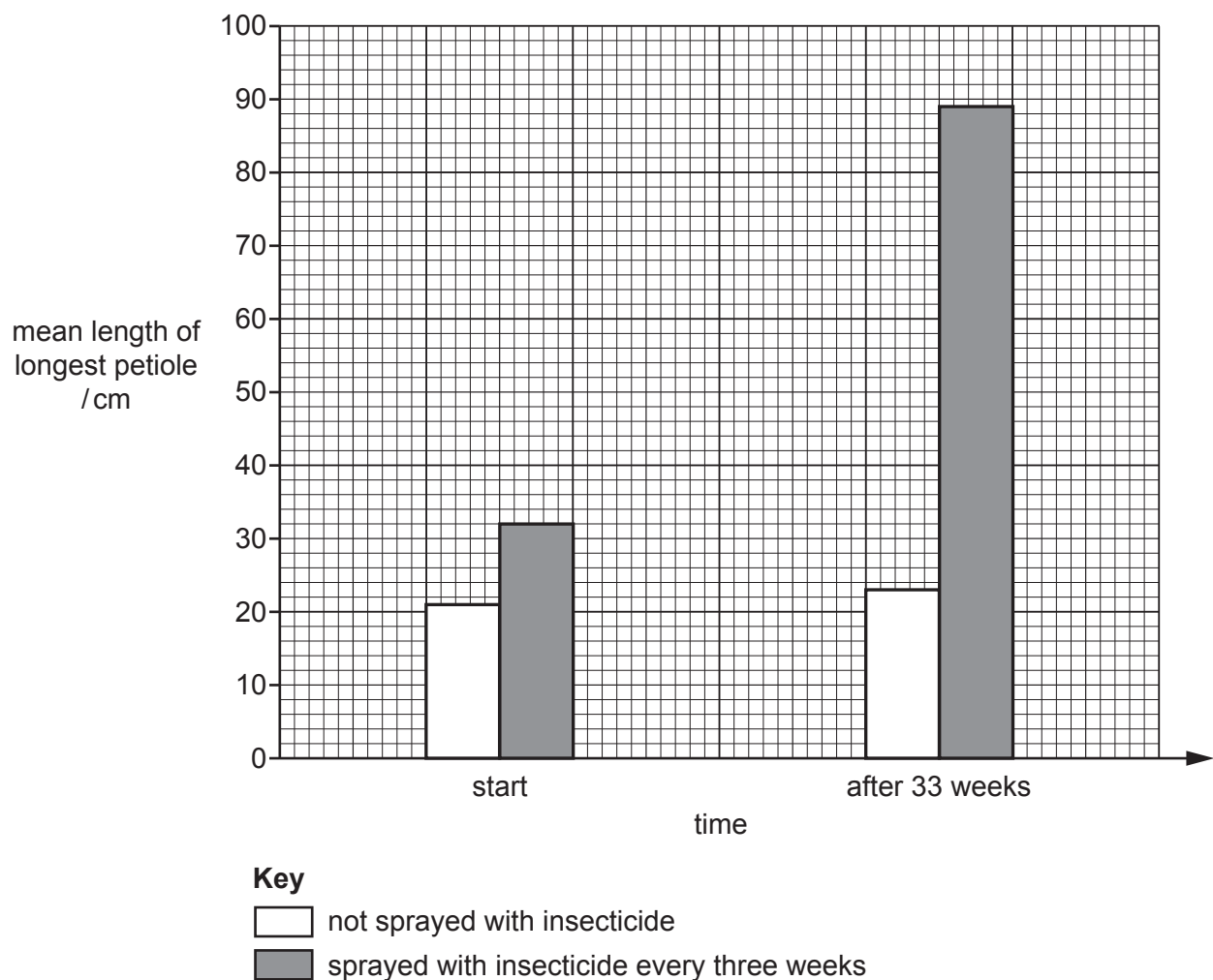


Fig. 2.1

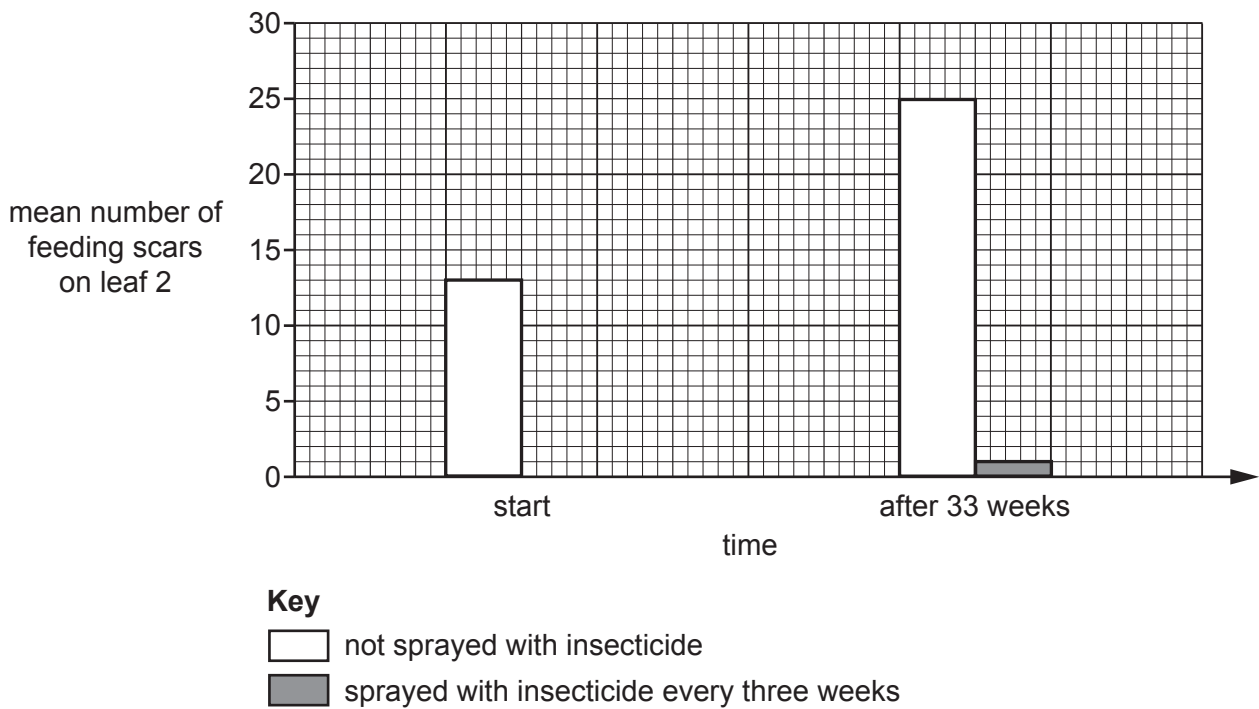


Fig. 2.2

- (b) (i) Calculate the percentage change in the mean length of the longest petiole for leaves sprayed with insecticide.

Show your working and give your answer to the nearest whole number.

.....% [2]

- (ii) State **one** reason why the *t*-test can be used to analyse the data in Fig. 2.2.

..... [1]

- (iii) State what information is needed to calculate the degrees of freedom used to determine if the calculated *t*-value is significant.

..... [1]

[Total: 11]